

SPRINT 03 · DELIVERABLE 04

AI Toolkit Hub.

*A unified platform for translation, summarization,
flashcards, voice translation, and text-to-speech.*

THE TEAM

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The logo features the letters 'AI' in a bold, black, serif font, centered within a solid cyan circle. This central circle is surrounded by four concentric, thin, light-blue circles, creating a ripple effect. The entire graphic is set against a dark blue background that also features a vertical cyan line on the left side of the slide.

AI

Agenda

Where we're going in the next 30 minutes

- 01** Sprint scope and deliverables
- 02** Project overview
- 03** Team roles and working agreement
- 04** Professor feedback improvements
- 05** Sprint 2 recap and Sprint 3 backlog
- 06** Completed and incomplete work
- 07** Testing, metrics, and retrospective
- 08** Sprint 3 plan and project demo

KEY FEATURES

Translation

Summarization

PDF → Flashcards

Voice Translation

Text-to-Speech

Problem Statement

Target users, pain points, and our solution hypothesis

WHO

Students, educators, professionals

Multilingual users working with long texts, PDFs, and cross-language communication.

WHAT

Tool fatigue and context loss

Time wasted switching between multiple tools for translation, summarizing, studying, and voice work.

WHY CURRENT SOLUTIONS FAIL

- Fragmented experience across 3–5 apps
- Limited history and export options
- Voice translation doesn't pair with text records

HYPOTHESIS

A single AI Toolkit Hub for students and multilingual professionals will reduce tool-switching time and improve comprehension by combining language and study workflows in one place.

Team Roles & Responsibilities

Ownership across product, frontend, backend, AI/ML, and QA / DevOps

Product Owner

Aryan Patel

Requirements, user stories, backlog priority, sprint planning

Frontend Lead

Bansari Shah

React.js UI, Tailwind CSS, API integration, flashcard UI

Backend Lead

Pratham Patel

Node.js APIs, JWT authentication, database, endpoints

AI / ML Lead

Vivek Pitroda

AI API integration, prompt design, PDF→flashcard logic

QA / DevOps

Lakshmi Prasanna

Testing, GitHub workflow, deployment, documentation

Team Working Agreement

Communication norms, meetings, and accountability

COMMUNICATION

- Google Meet, WhatsApp, Notion
- 24-hour response time
- Early notice when unavailable

MEETINGS

- Saturday 4–5 PM (Virtual)
- Monday 9–10 PM (Virtual)
- Wednesday 4:30–5:30 PM (In-person)

WORK EXPECTATIONS

- Complete tasks by deadline
- Reach out early if blocked
- Fair workload distribution

ACCOUNTABILITY & QUALITY

- Supportive accountability
- Open about missed deadlines
- Review work before submission

Improvements from Professor Feedback

Direct response to last sprint's review

01

Re-scoped MVP

Now includes usable product features — not authentication alone.

02

Sprint recap added

Clear Sprint 2 recap and Sprint 3 summary sections introduced.

03

Aligned artifacts

Backlog, test cases, and metrics are aligned with the current sprint.

Liam Chen

Language Learner / Accessibility User



Liam Chen

Language Learner / Accessibility User

GOALS

- Practice real-time conversations via voice translation
- Generate audio from text content
- Improve language comprehension

HOW AI TOOLKIT HELPS

- Live voice translation supports conversation practice
- Text-to-speech improves accessibility
- Translator supports multilingual learning

FRUSTRATIONS

No real-time support in many apps · Too many separate tools

Daniel Carter

Busy Professional



Daniel Carter

Busy Professional

GOALS

- Translate client communication quickly
- Summarize long reports into key points
- Listen to content while multitasking

HOW AI TOOLKIT HELPS

- Translator speeds communication
- Summarizer extracts main insights
- Text-to-speech supports multitasking

FRUSTRATIONS

Reading long reports is time-consuming · Using multiple tools slows work

Maria Gonzalez

International Student



Maria Gonzalez

International Student

GOALS

- Translate academic material quickly
- Summarize long articles for study
- Convert PDFs into flashcards

HOW AI TOOLKIT HELPS

- Translation helps understand course material
- Summaries reduce reading time
- PDF-to-flashcards creates study support

FRUSTRATIONS

Complex English content is difficult · Manual study-note creation takes time

Minimum Viable Product

The smallest usable version of AI Toolkit Hub for Sprint 3

Authenticated users can interact with the product through voice-based features inside a single, unified application.

01

User registration & login

02

Complete UI flow

03

Backend API integration

04

Voice translation feature

05

Base dashboard / navigation

06

Text-to-Speech (TTS)

Technologies

Frontend, backend, AI services, and database

FRONTEND

React.js · Tailwind CSS · Axios · react-card-flip · Framer Motion

BACKEND

Node.js + Express · Multer (uploads) · pdf-parse · JWT + bcrypt

AI SERVICES

MyMemory Translation API · Hugging Face (BART, FLAN-T5) · Web Speech API

DATABASE

MongoDB

Algorithms & Processing

How each feature is powered

1

Text Translation

MyMemory API + rule-based handling

Input validation, language mapping, history logging

2

Text Summarization

facebook/bart-large-cnn

Length control, formatting, response cleanup

3

PDF → Flashcards

google/flan-t5-large + pdf-parse

Extraction, chunking, Q&A generation, JSON formatting

4

Live Voice Translation

SpeechRecognition → Translate → Synth

Real-time transcription, API chaining, fallback

5

Text-to-Speech

Web Speech API

Voice selection, pitch / rate control, playback

6

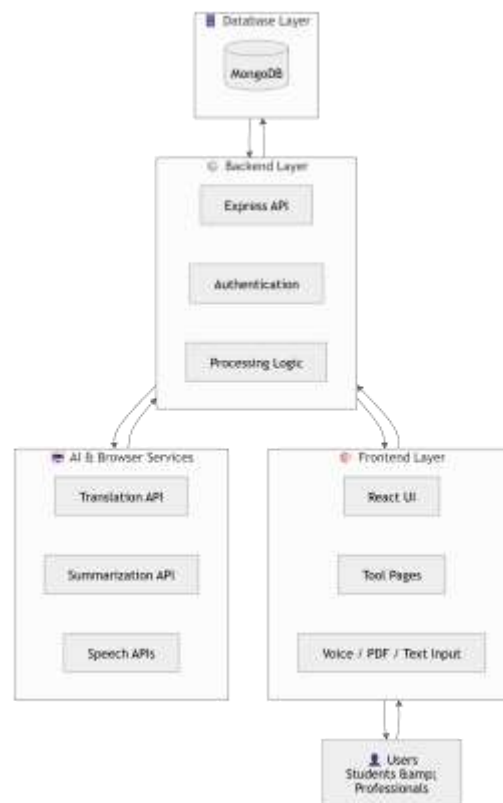
Authentication

JWT + bcrypt

Password hashing, token gen, middleware verification

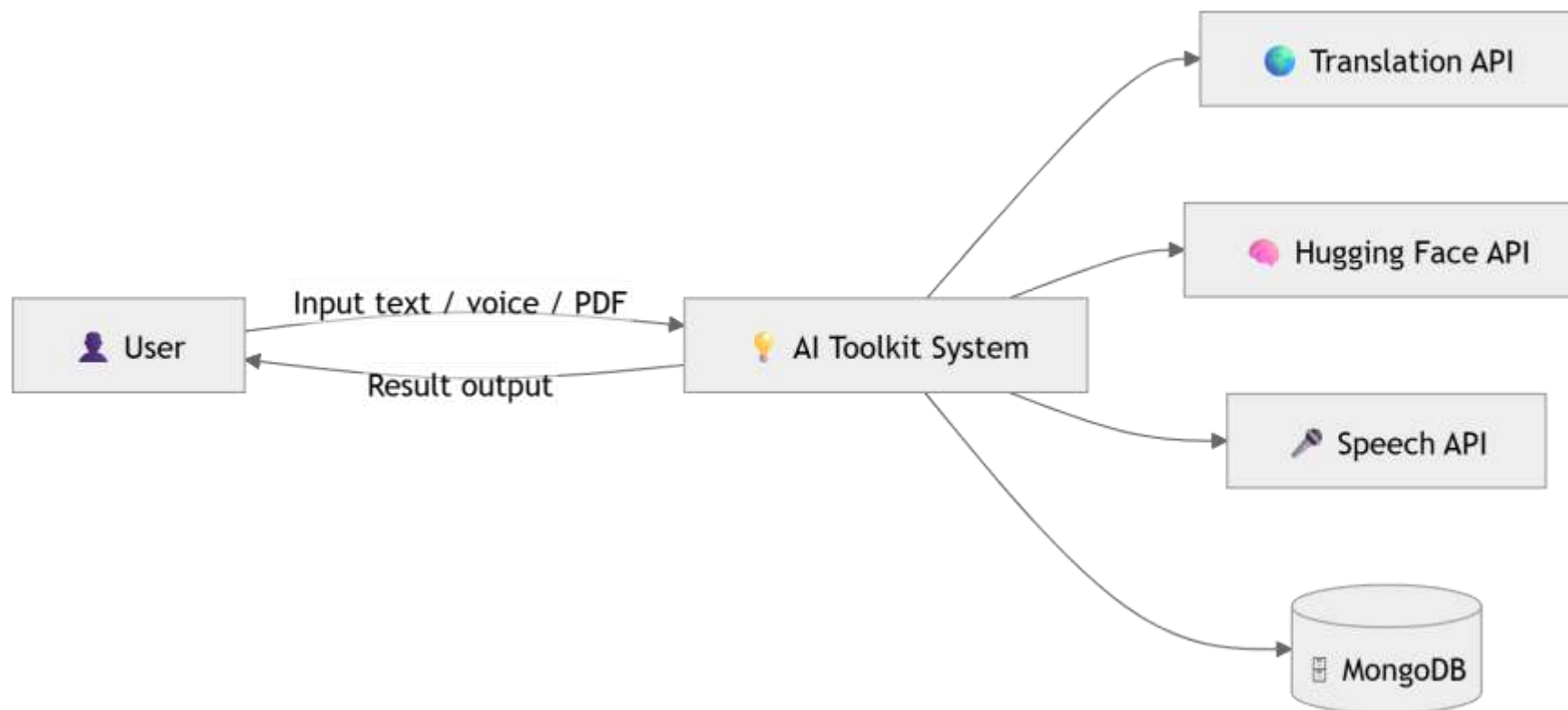
System Architecture

Updated architecture diagram



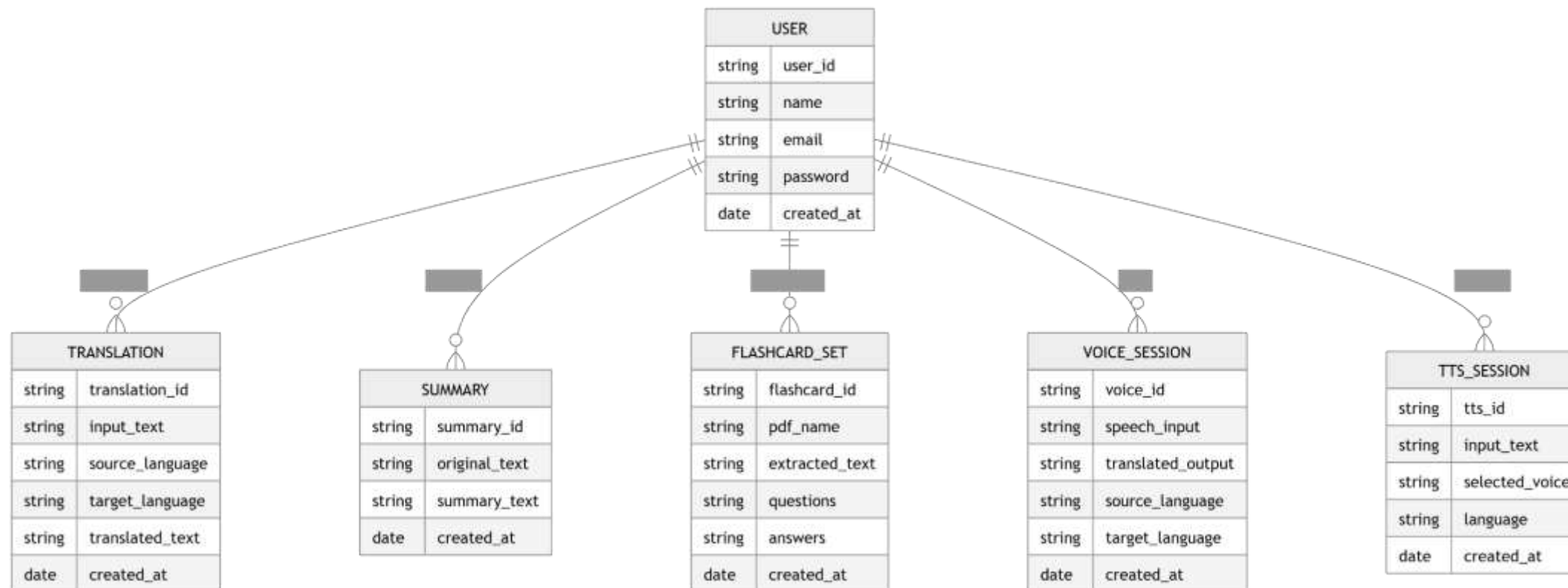
Context Diagram

High-level system context with users and services



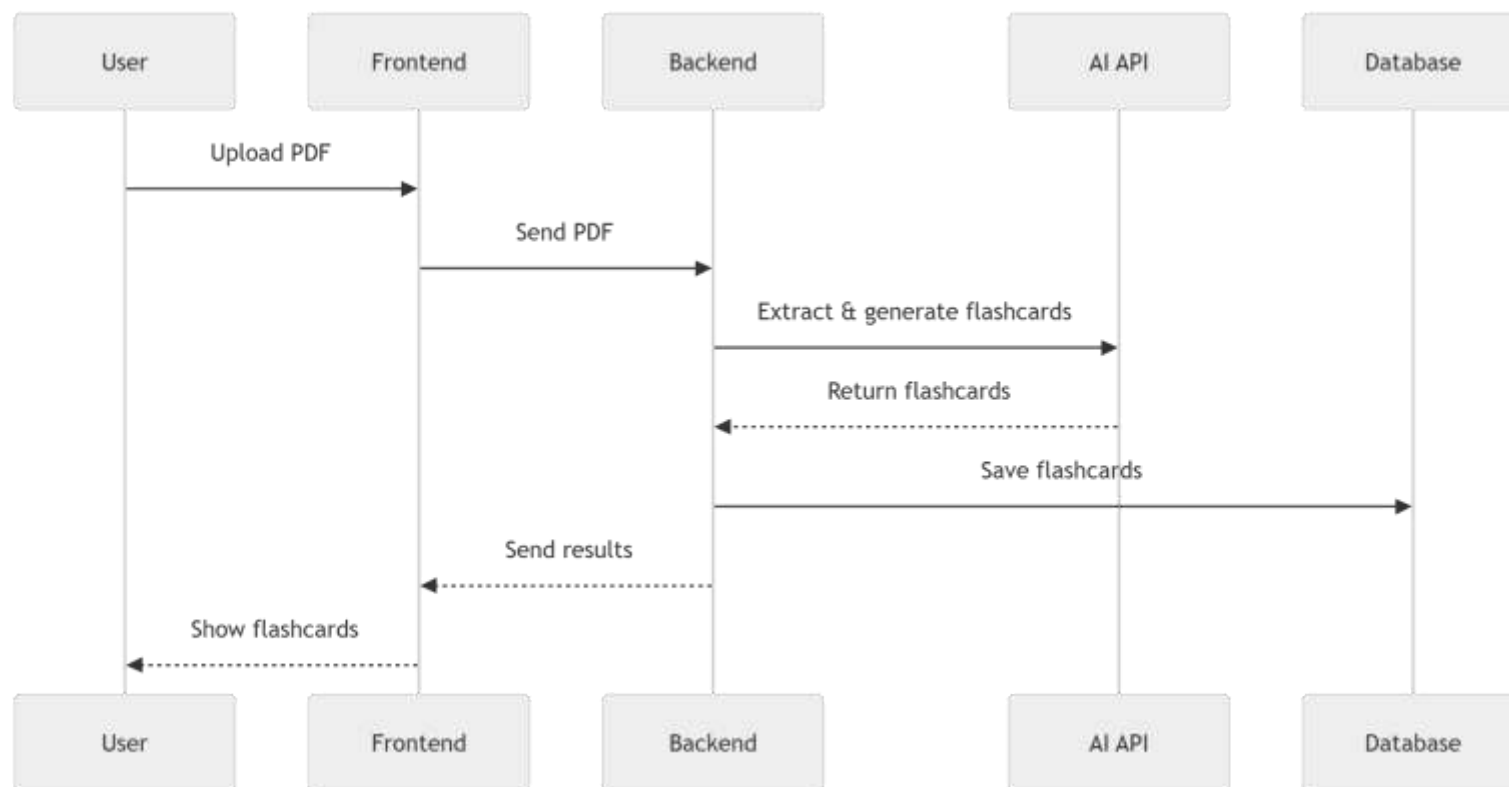
ER Diagram

Updated database design and entity relationships



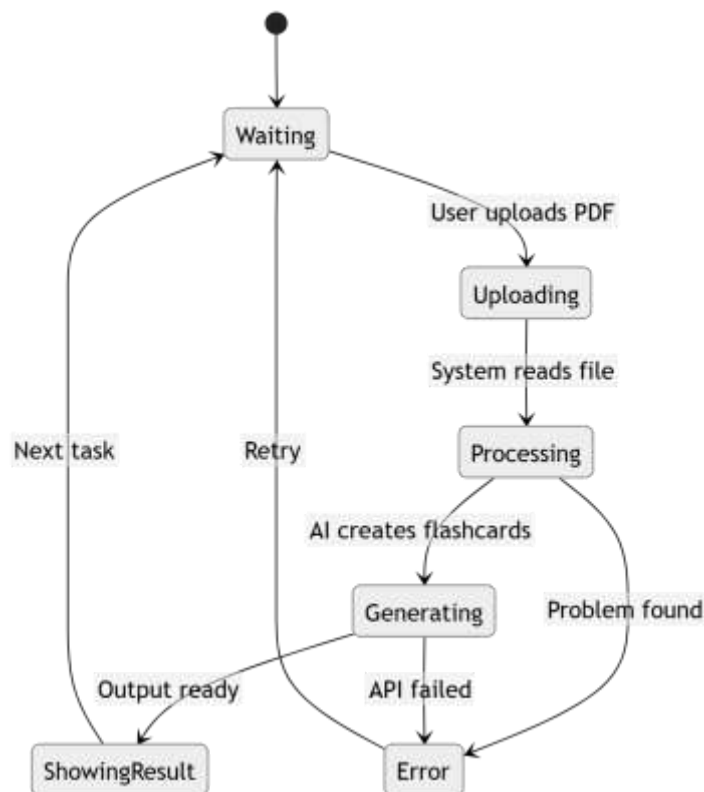
Sequence Diagram

Interaction flow for the PDF-to-flashcards feature



State Diagram

State transitions for flashcard generation



Sprint 2 → Sprint 3

What we built on, and where we took it

WHAT SPRINT 2 ESTABLISHED

- Authentication foundation — login + registration
- Dashboard structure and MongoDB connection
- Initial API testing using Postman
- Base architecture and diagrams prepared

HOW SPRINT 3 BUILDS ON IT

- Transitioned UI toward an interactive, voice-enabled experience
- Real-time voice translation across multiple languages
- Text-to-Speech (TTS) for dynamic audio responses
- Speech-to-Text (STT) for seamless voice input
- Shifted focus from setup to intelligent, user-centric features

Sprint Journey

Four deliverables, one product



DELIVERABLE 1

Sprint 0 — Foundation



Planning, architecture, auth setup,
dashboard basics

DELIVERABLE 2

Sprint 1 — Progress



Improved flows, diagrams, feedback
updates, core integration

DELIVERABLE 3

Sprint 2 — Progress



UI completed, all APIs completed,
flashcard generation done

DELIVERABLE 4

Sprint 3 — Current



UI complete, all APIs complete, voice
translation, TTS done

Sprint 3 Backlog

Committed work for the current sprint

ID	STORY / TASK	STATUS
S3-01	Complete application UI across main screens	DONE
S3-02	Finish all backend APIs and integrations	DONE
S3-03	Implement voice translation functionality	DONE
S3-04	Integrate Text-to-Speech (TTS) for audio output	DONE
S3-05	Build profile page UI	DONE

User Stories — Overview

Completed stories and carryover

COMPLETED

- Fully functional UI — seamless navigation across core screens
- Backend APIs fully integrated and validated
- Real-time voice translation enabling multi-language communication
- Secure authentication and profile management implemented
- Text-to-Speech (TTS) delivering dynamic audio responses

100%

Sprint 3 delivery

No carryover items

User Stories & Acceptance

Authentication and Voice Translation

U S - 1

Authentication Access

As a user, I want to register and log in securely so that I can access the AI Toolkit Hub and my personalized workspace.

ACCEPTANCE

- ✓ Account creation with required fields
- ✓ Login with valid credentials
- ✓ Invalid credentials show an error
- ✓ Protected screens require auth

U S - 2

Voice Translation

As a user, I want to translate spoken or typed input into another language so I can communicate easily.

ACCEPTANCE

- ✓ Voice or text input accepted
- ✓ Speech-to-Text converts accurately
- ✓ Translated output in selected language
- ✓ Multiple languages supported

User Stories & Acceptance

Text-to-Speech and Product UI

U S - 3

Text-to-Speech

As a user, I want to hear translated or input text as audio so I can understand pronunciation and improve accessibility.

ACCEPTANCE

- ✓ Clear audio output from text
- ✓ Smooth, responsive playback
- ✓ Multiple languages and voices
- ✓ Works on translated and original text

U S - 4

Complete Product UI

As a user, I want a clean, navigable interface so I can easily use all voice features.

ACCEPTANCE

- ✓ Main screens load correctly
- ✓ Smooth navigation across features
- ✓ Clean, usable release demo
- ✓ Visually complete for presentation

Sprint 3 Test Cases

Coverage across authentication, navigation, integration, and voice

ID	Area	Test Action	Expected	Status
TC-01	Authentication	Register with valid user data	Account created successfully	✓
TC-02	Authentication	Login with registered credentials	User redirected into app	✓
TC-03	UI Navigation	Navigate across core app screens	Seamless navigation, no errors	✓
TC-04	API Integration	Trigger backend APIs via frontend	Accurate, timely responses	✓
TC-05	Voice Translation	Input voice / text and translate	Real-time, multi-language result	✓
TC-06	Text-to-Speech	Generate audio from input / translated text	Clear, responsive audio output	✓

Completed in Sprint 3

Everything we shipped this sprint



UI Complete

All main screens delivered with polish and consistency



All APIs Complete

Backend endpoints integrated and validated end-to-end



Flashcard Generation

PDF → flashcards working through FLAN-T5 pipeline



Auth: Login & Register

Secure JWT-based authentication flow stable

Sprint 3 Metrics

Best-effort using current task count

6

COMMITTED TASKS

6

COMPLETED TASKS

100%

COMPLETED / COMMITTED

DELIVERED

UI · APIs · Voice Translation · TTS · Auth · Profile

Team Velocity

This sprint vs. historical average

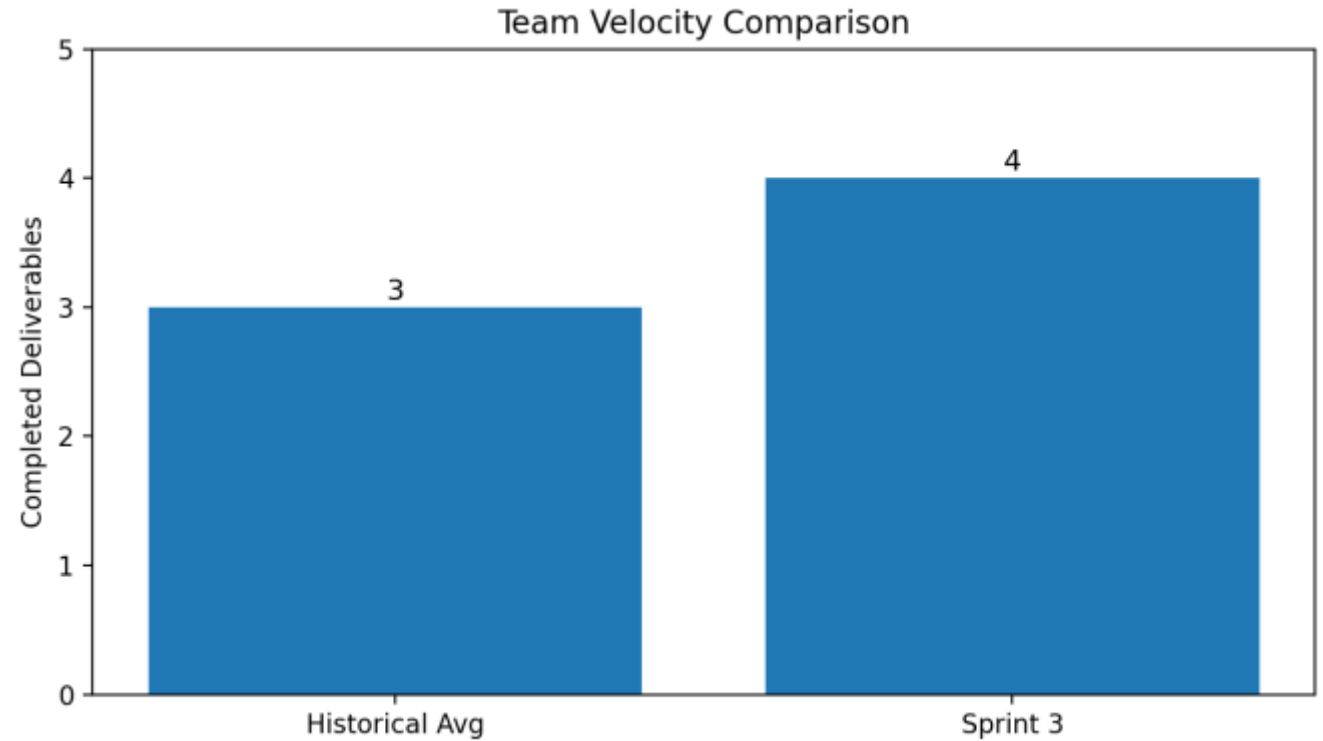
THIS SPRINT

6

completed deliverables

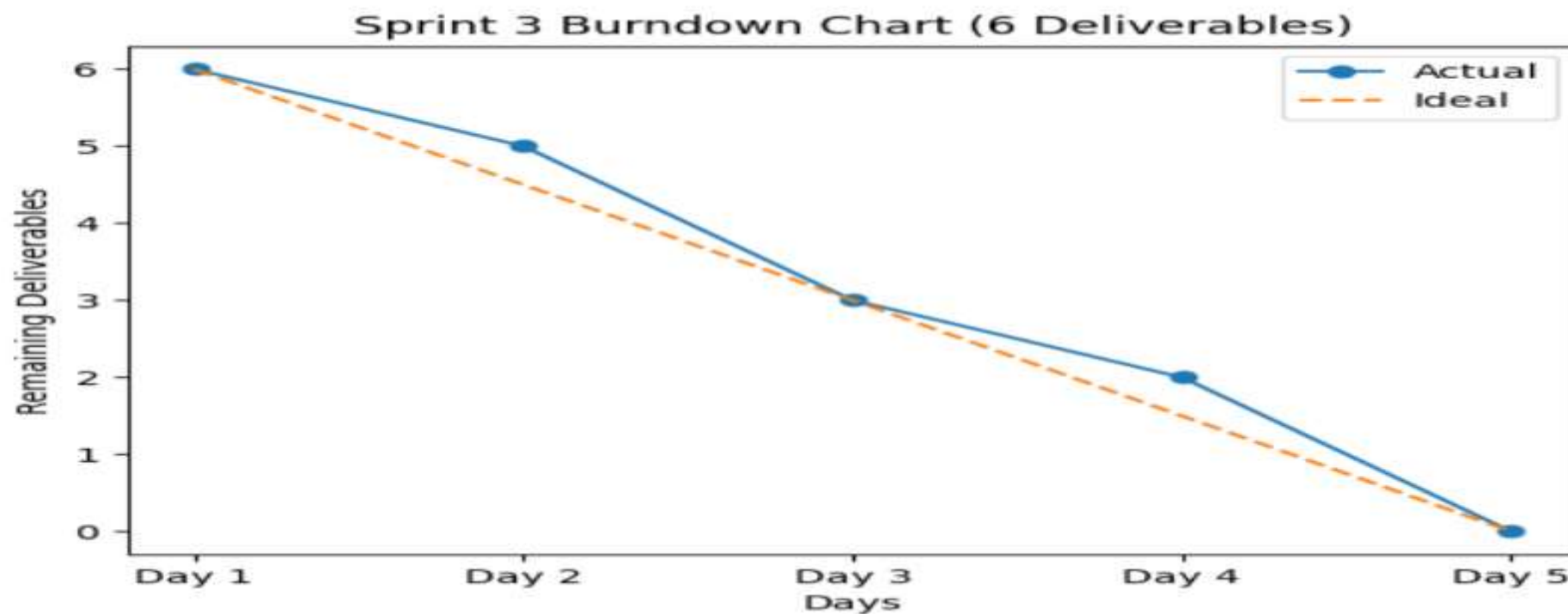
HISTORICAL AVG

3 deliverables / sprint



Burndown Chart

Tasks remaining over the sprint



Sprint Retrospective

What went well and what to improve

WHAT WENT WELL

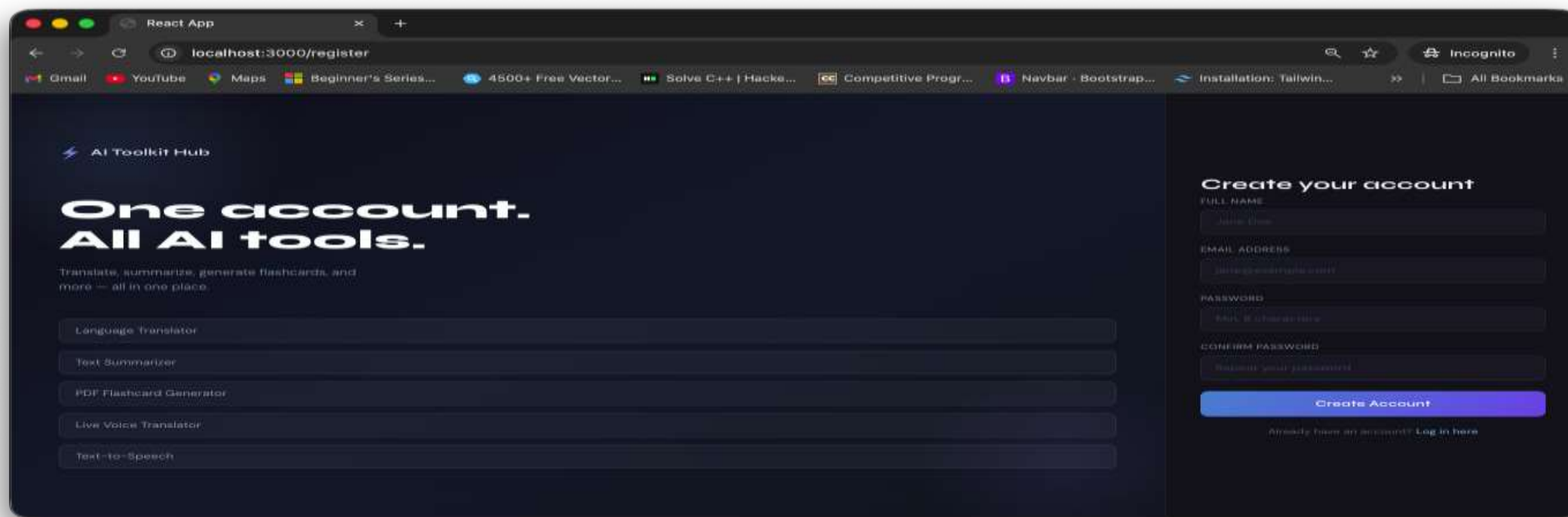
- ✓ Main UI implementation completed successfully
- ✓ All APIs fully integrated and working
- ✓ Voice translation implemented successfully
- ✓ Text-to-Speech with smooth audio output
- ✓ Core authentication and profile are stable

WHAT TO IMPROVE NEXT

- No major improvements identified — sprint executed as planned
- Continue maintaining code quality and performance standards
- Explore enhancements and new feature opportunities next sprint

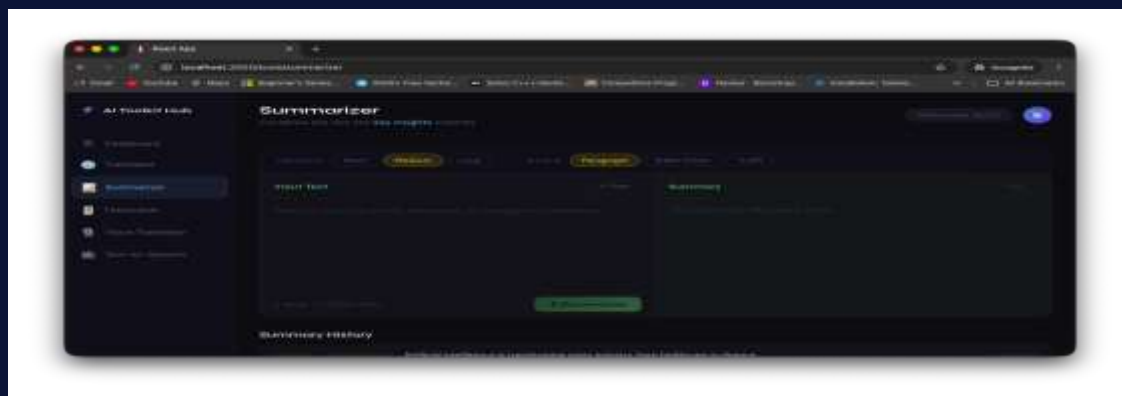
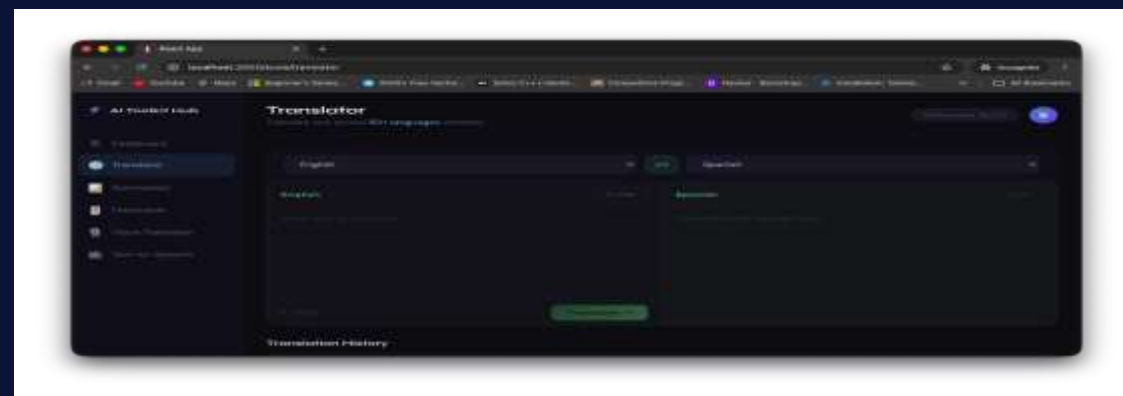
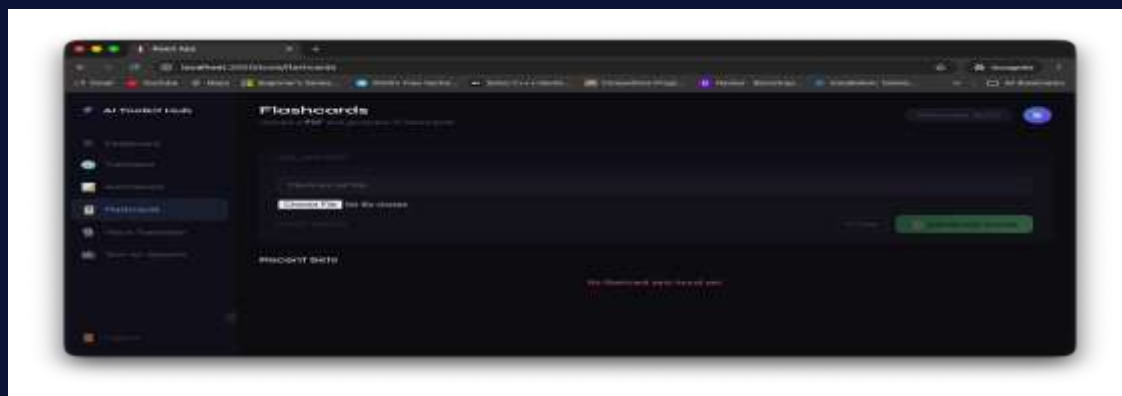
Project Demo

Authentication entry point — modern React UI



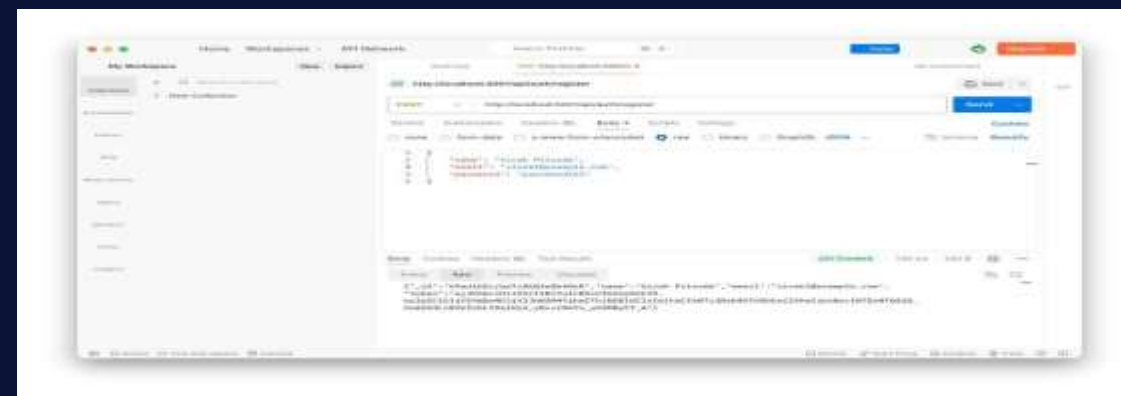
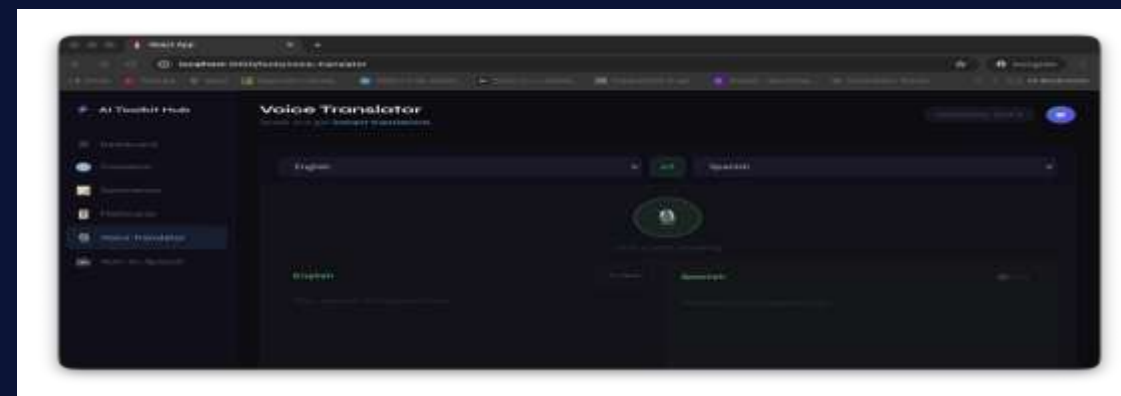
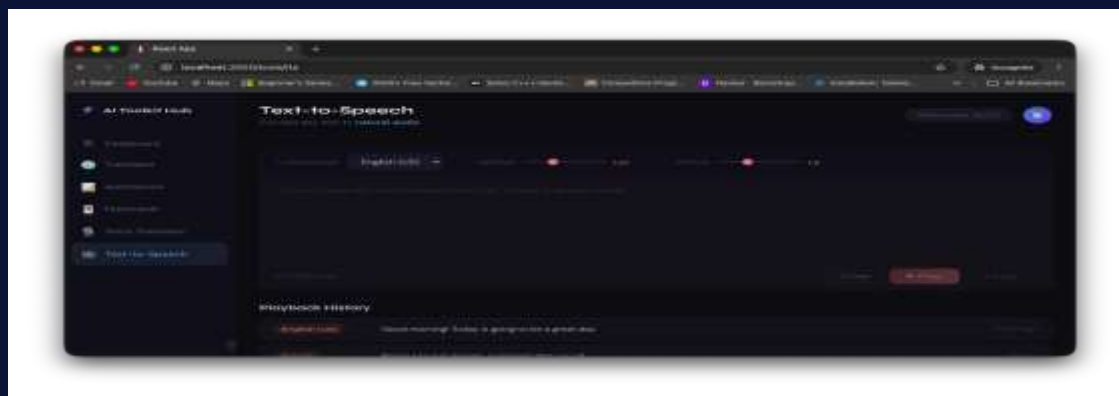
Inside the Product

Core screens across the application



Voice & Translation in Action

Live voice features and translation flows



Q & A · END OF DELIVERABLE 4

Thank you.

AI Toolkit Hub — Sprint 3 Presentation

Aryan · Bansari · Lakshmi · Pratham · Vivek

A light blue circle containing the letters "AI" in a dark blue, serif font.

AI